

Deployment of Web-based Immersive VR in a Korean Elementary Classroom: A Field Study on Earth and Moon Movements

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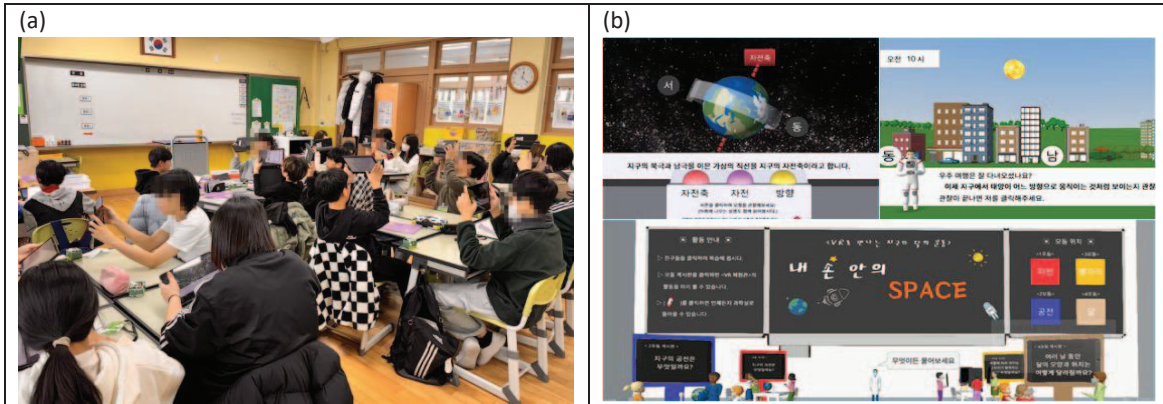


Figure 1: The web-based immersive VR learning environment deployed in a Korean elementary classroom. (a) Students complete guided tasks and interactions on tablets; (b) students use Google Cardboard for stereoscopic observation of 3D scenes.

ABSTRACT

Teaching abstract concepts like Earth and Moon movements is challenging in K-12 education due to the difficulty of observing celestial phenomena directly. To address this, we integrated a web-based VR curriculum, which was initially developed and recognized for its educational efficacy in a national research project [7], into a Korean elementary science classroom, combining tablet-based interactions with stereoscopic viewing via a smartphone-based HMD (Google Cardboard). A field study with 48 sixth-grade students compared this VR approach against traditional instruction. Results indicate the VR group achieved statistically significant gains in scientific attitude ($t = 2.61, p = .012$) and self-directed learning (Q1–Q5; all items $p < .05$). This study demonstrates the practical viability and educational effectiveness of deploying accessible VR courseware in real-world school settings.

Index terms: Immersive VR, Elementary Science Education, Field Study, Interactive Learning Environment, Earth and Moon Observation.

1 INTRODUCTION

Based on experiential learning theory, active engagement in 3D environments is crucial for understanding abstract scientific concepts [4]. In South Korea, the 2015 National Curriculum emphasizes student-centered activities to foster core competencies [6]. However, the "Earth and Space" domain in elementary science presents unique challenges for students transitioning from concrete to formal operational stages [5]. Virtual reality (VR) learning

environments have long been recognized for their potential to transcend physical limitations [2] and transform the social context of learning for both teachers and students [1]. To address these challenges, this study proposes a web-based immersive VR courseware tailored for the Korean elementary science curriculum. We deployed this system in a real-world classroom and verified its effectiveness through a comparative experiment.

2 SYSTEM DESIGN

We developed the courseware using Delightex Edu to ensure classroom accessibility on tablets while supporting stereoscopic viewing via smartphone-based HMDs (e.g., Google Cardboard) for immersive 3D observation [3].

2.1 Content Development

The content aligns with the 6th-grade unit "Movement of the Earth and Moon" and consists of two main modules. First, the Operation-Oriented Module allows students to observe the Earth from an exocentric viewpoint. By rotating the Earth and observing the sun's relative position, students can intuitively understand the cause of day and night. Second, the Storytelling-Based Module offers a gamified narrative where students solve quizzes about constellations and moon phases, simulating observation conditions impossible in reality.

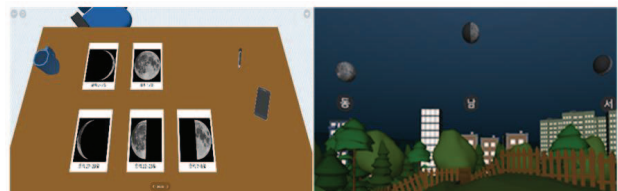


Figure 2: The storytelling-based module where students simulate moon phases by manipulating the time flow.

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3 EXPERIMENT

To verify educational effectiveness, we conducted a field study at an elementary school in Daegu, South Korea.

3.1 Participants

A total of 48 sixth-grade students participated in the study (Experimental $N = 24$, Control $N = 24$). The Control Group used traditional textbook-based audio-visual materials, while the Experimental Group used the developed VR courseware with tablet-based interactions and Google Cardboard for stereoscopic observation. A homogeneity test confirmed no significant difference in prior achievement ($p = .743$) or attitude ($p = .864$) between the two groups.

3.2 Procedure

The experimental procedure is summarized in Figure 3. Both groups first took a pre-test for basic academic diagnosis and Scientific Attitude. During the intervention phase, both groups studied the same unit ("Movement of the Earth and Moon"). The Control Group received traditional instruction utilizing standard textbooks and 2D video materials. In contrast, the Experimental Group interacted with the VR content on tablets (e.g., guided exploration and task completion). For stereoscopic viewing, students additionally used Google Cardboard to observe 3D Earth-Moon phenomena in an immersive manner. Finally, a post-test measured Scientific Attitude and self-directed learning using validated 5-point Likert questionnaires (Attitude: Cronbach's $\alpha = 0.91$; SDL: 5 items, Q1-Q5).

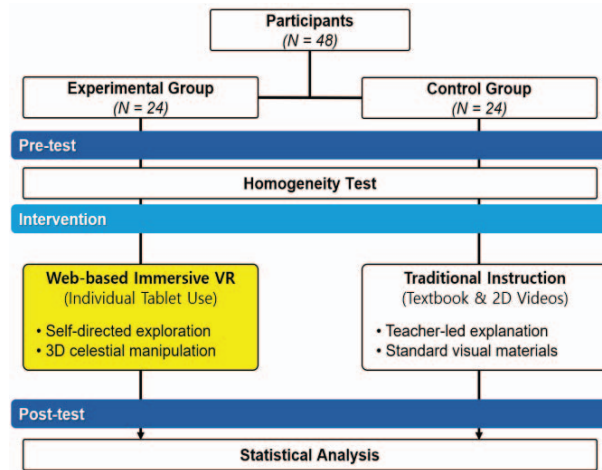


Figure 3: The workflow of the developed VR courseware.

4 RESULTS

We conducted an independent-samples t-test to compare the post-test scientific attitude scores between the two groups. Table 1 shows the statistical results.

Table 1. Results of Post-test on Scientific Attitude					
Groups	N	M	SD	t-value	p-value
Control	24	4.04	0.69	2.61	.012
Experimental	24	4.50	0.51		

As shown in Table 1 and Figure 4, the experimental group ($M = 4.50$, $SD = 0.51$) showed higher scientific attitude scores than the

control group ($M = 4.04$, $SD = 0.69$). The t-test result ($t = 2.61$, $p = .012$, *Cohen's d* = 0.76) indicates that this difference is statistically significant. This suggests that the VR learning experience was associated with more positive attitudes toward science.

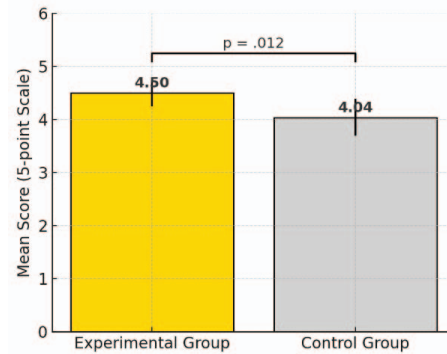


Figure 4: Comparison of post-test scientific attitude scores. The VR group showed significantly higher scores ($p = .012$).

Furthermore, self-directed learning was higher in the VR group than in the control group (Q1-Q5; all items $p < .05$; e.g., Q4: VR $M = 3.7$, $SD = 0.62$ vs. Control $M = 2.9$, $SD = 0.50$; $|t| = 4.83$).

5 CONCLUSION

This study suggests that web-based immersive VR can be an effective tool for teaching abstract scientific concepts in K-12 education. By deploying the system in a real Korean elementary classroom, we confirmed that students using VR achieved higher scientific attitude scores compared to traditional methods. While this study confirms the effectiveness of web-based VR, future research should address limitations such as the novelty effect and the small sample size from a single school.

ACKNOWLEDGMENTS

This work was supported by the Institute of Information & Communications Technology Planning & Evaluation(IITP)-Innovative Human Resource Development for Local Intellectualization program grant funded by the Korea government(MSIT)(IITP-2026-RS-2022-00156389).

REFERENCES

- [1] J. N. Bailenson, N. Yee, J. Blascovich, A. C. Beall, N. Lundblad, and M. Jin, "The use of immersive virtual reality in the learning sciences: Digital transformations of teachers, students, and social context," *The Journal of the Learning Sciences*, vol. 17, no. 1, pp. 102-141, 2008.
- [2] M. Bricken, "Virtual reality learning environments: Potentials and challenges," *SIGGRAPH Computer Graphics*, vol. 25, no. 3, pp. 178-184, 1991.
- [3] Delightex, "CoSpaces Edu: Make AR & VR in the classroom," [Online]. Available: <https://edu.delightex.com/BWV-FHG>. [Accessed: Dec. 01, 2025].
- [4] J. Fromm, J. Radianti, C. Wehking, S. Stieglitz, T. A. Majchrzak, and J. vom Brocke, "More than experience? On the unique opportunities of virtual reality to afford a holistic experiential learning cycle," *The Internet and Higher Education*, vol. 50, p. 100804, 2021.
- [5] W. Huitt and J. Hummel, "Piaget's theory of cognitive development," *Educational Psychology Interactive*, vol. 3, no. 2, pp. 1-5, 2003.
- [6] Ministry of Education, 2015 National Curriculum, Sejong: Ministry of Education, 2015.
- [7] S. Ryu and T. Yang, "Earth and Moon Movements through VR. In Proc. 15th Nat. Education Information Research Contest, Daegu, Korea, 2021.